Phone Thant Kyaw

Computer Engineering — machine learning, computer vision, and applied systems

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Summary

Fourth-year Computer Engineering student at Mae Fah Luang University (GPA 3.98/4.00, TOEIC 980/990) and primary developer of a roadside accident-risk warning prototype that pairs YOLO-based detection with a trained classifier. Works across dataset preparation, model training, and integration, with further experience in web, mobile, and hardware prototyping. Available for a 16-week co-op internship, 4 Jan – 23 Apr 2027; dates flexible.

Education

Mae Fah Luang University — B.Eng. in Computer Engineering Chiang Rai, Thailand
Aug 2023 – Aug 2027 (expected)

- **GPA 3.98/4.00** | **TOEIC Listening & Reading 980/990** (CEFR C1, Aug 2026) | Taught in English
- **Relevant coursework:** Machine Learning; Computer Vision and Visual Analytics; Big Data Management and Analytics; Data Structures and Algorithms; Web and Mobile Application Development; Data Communications and Networks; Embedded Systems and IoT

Experience

Teaching Assistant (Laboratory) Chiang Rai, Thailand
Mae Fah Luang University Aug 2025 – Present

- Support laboratory teaching across five courses spanning programming, data communications and networks, digital logic and microcontrollers, and data visualisation and analytics.
- Troubleshoot student programming, software, and hardware problems in 10+ lab sessions per semester, each with 50+ students.
- Prepare and revise lab materials, grade coursework, and help instructors build grading rubrics and assess final group projects across three semesters.

Selected Projects

Real-Time Smart Road Accident Risk Warning Senior Project, Jan 2026 – Present
Project Manager & Primary Developer | Python, YOLO11, LightGBM, OpenCV, Docker

- Lead a three-person team building a roadside prototype that classifies accident-risk type from traffic video and displays a graded warning; responsible for most of the dataset, modelling, and integration work.
- Prepared and feature-engineered 139,924 Thai road accident records (2019–2025) across five heavily imbalanced classes; compared LightGBM, XGBoost, CatBoost, and random forest against oversampling, undersampling, and SMOTE, selecting LightGBM with random oversampling at **0.7582 test macro recall** (macro recall chosen because the rare classes matter most).
- Integrated the selected LightGBM pipeline into a YOLO and OpenCV application, adding road-region filtering, motorcycle-rider exclusion, and pedestrian-presence validation to suppress out-of-scope detections and incorrect pedestrian warnings.
- Added confidence-banded warnings and CSV event logging, verified end to end on recorded footage.

Aviation Flight Delay Analytics and Alert System Mar – May 2026
Python, PySpark, Spark MLlib, Apache Kafka

- Owned model training, evaluation, and testing on 454,414 cleaned 2023 flight records, comparing logistic regression, decision trees, random forests, and gradient-boosted trees in Spark MLlib.
- Selected gradient-boosted trees with oversampling, using PR-AUC as the primary metric because delayed flights were the minority class; tested the team's Kafka streaming prediction prototype.

University Room Reservation System (Web & Mobile) 2025
Node.js, Express, MySQL, Flutter | Web repo · Mobile repo

- One of the primary developers of a room-booking platform with reservation requests, a lecturer approval workflow, and role-based access for students, lecturers, and staff.
- Built and tested the Express and MySQL REST API with JWT authentication and argon2id password hashing, and extended the system into a Flutter mobile app tested on the iOS simulator.

Ka-Naung Pet Dog — Robotic Hardware Prototype Jul 2026 – Present
Hardware development and testing | collaborative team project

- Test MG995 servo motors, evaluate component power draw against battery options, and contribute soldering, assembly, and PCB and remote-control connection planning.

Technical Skills

- **Programming:** Python, JavaScript, Dart, SQL, HTML/CSS; coursework in Java, C#, MATLAB
- **Machine learning & data:** LightGBM, XGBoost, CatBoost, scikit-learn, YOLO, OpenCV, PySpark, Spark MLlib, Kafka
- **Software & tools:** Node.js, Express, Flutter, MySQL, REST APIs, Docker, Git, Linux
- **Hardware:** Soldering, servo motor testing, power-budget and battery assessment, microcontroller fundamentals

Additional Information

- **International exchange:** FACE IT 2026 — Deep Learning and AI for Images, Universitas Brawijaya (FILKOM), Indonesia, Sep 2026
- **Online courses:** Getting Started with AWS Machine Learning; HPE Aruba Networking Basics
- **Languages:** Burmese (native); English (TOEIC 980/990); Thai (beginner); Japanese (N5 level)